Vaibhav Tiwari

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Summary

Software Development Engineer with 2+ years of experience building enterprise applications and distributed services using Java, Spring Boot, Microservices, REST APIs, and AWS. Experienced in designing scalable backend systems, performance optimization, application modernization, with hands-on experience in Kafka, Redis, PostgreSQL, Docker, and distributed locking. Strong problem-solving skills with a 1900+ LeetCode rating.

Technical Skills

Languages: Java, C++, JavaScript, SQL

Backend: Spring Boot, Spring Data JPA, Hibernate, REST APIs, Microservices

Distributed Systems: Kafka, Redis, Distributed Locking, Event-Driven Architecture, Caching

Databases: PostgreSQL, MongoDB, Redis

Cloud & DevOps: AWS (EC2, S3, RDS, Lambda, CloudWatch), Docker, GitHub Actions, CI/CD

Testing: JUnit

Frontend: React.js, React-Redux

Experience

System Engineer Feb 2025 – Present
Tata Consultancy Services (TCS) | Client : NatWest Group *Gurgaon, India*

- Enabled arrears visibility for **63% of bank users** on mobile by building a **REST API** aggregating data from **2 in-house applications**.
- Cut duplicate agent calls by **30%** via an **API** that blocks repeat outreach to recently contacted customers.
- Secured **100+ application credentials** and eliminated hardcoded secrets by integrating **HashiCorp Vault**, strengthening application security and centralized secret management.
- Modernized **6 Java Spring Boot microservices** by migrating them from Java 8 to Java 17, improving platform compatibility and reducing end-of-life risks across distributed backend services.
- Prevented **certificate expiry-related service disruptions** by proactively renewing **3 types of SSL/TLS certificates** across **7–8 microservices**, maintaining secure service-to-service communication.
- Reduced **tracker preparation time by 66%** by automating repetitive manual tracking tasks, improving process consistency and accuracy.

Associate Software Engineer Jul 2024 – Jan 2025
Innostax Software Labs Ltd. *Gurgaon, India*

- Reduced **API response time by 37%** by implementing caching strategies and optimizing database queries, improving backend performance and reducing latency in application workflows.
- Improved the application's **Lighthouse performance score from 50% to 93%** by optimizing frontend performance and page-load efficiency, enhancing overall user experience.
- Expanded booking capabilities by enabling **zero-cost bookings without payment processing**, implementing the **zero-payment booking flow** for eligible itineraries in the **Tavatrip flight-booking application**.
- Accelerated frontend development across key application workflows by engineering **4+ responsive React.js interfaces** from client-provided Figma designs using reusable components.

Projects

Eventura – Full Stack Microservices Ticket Booking Platform

- Designed a **distributed Spring Boot microservices architecture** comprising **7 independent services** with an API Gateway for authentication, event catalog, show management, booking, payments, and notification workflows.
- Prevented **concurrent seat allocation and double-booking** by implementing **Redis-based distributed locking** with TTL-based seat locks, ensuring consistency across concurrent booking requests.
- Enabled **asynchronous communication across 3 microservices** using **Kafka-based event-driven workflows**, with Redis caching and PostgreSQL persistence to support scalable service interactions.
- Established **independent service execution and reproducible development environments** by containerizing **7 Spring Boot microservices** using Docker.

Achievements & Problem Solving

LeetCode Knight 1900+ Rating | Top 3% Globally | Solved 250+ Problems

Education

Ajay Kumar Garg Engineering College **Nov 2020 – Jun 2024**
B.Tech in Computer Science and Engineering | CGPA: 7.69/10 *Ghaziabad, India*